Elastic On-Device LLM Service

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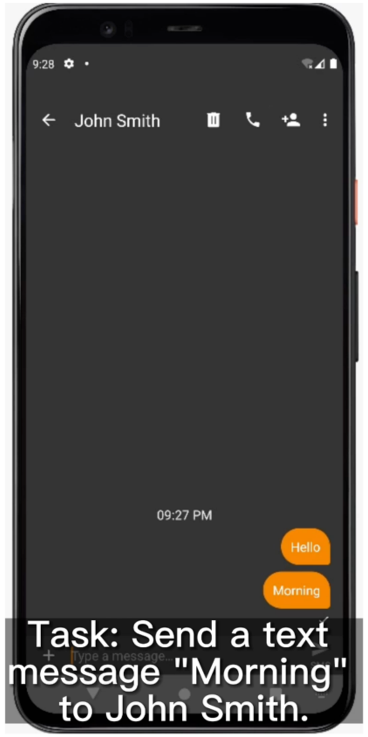
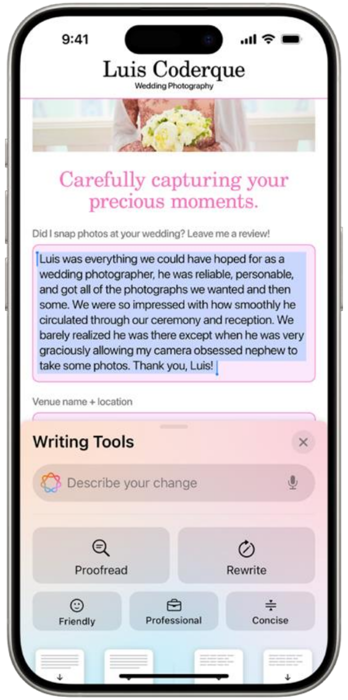
On-Device Large Language Models



Automatic GUI Manipulation



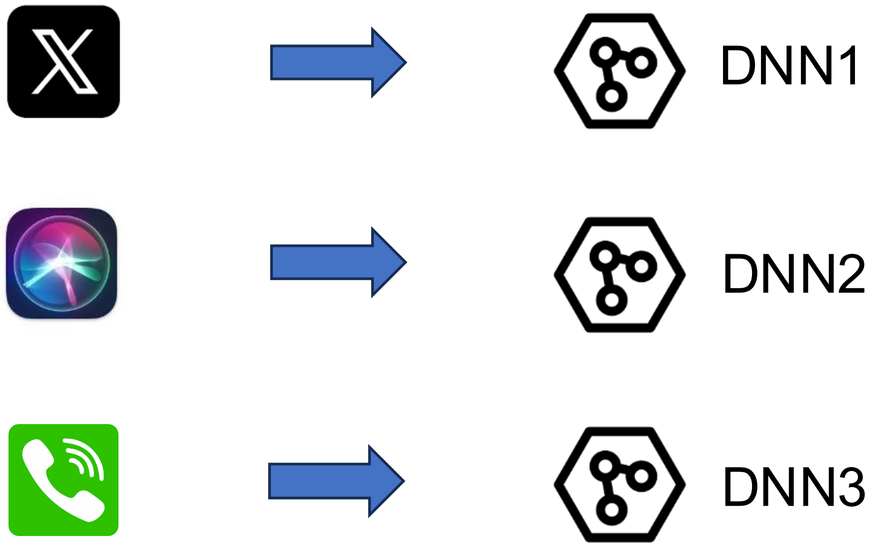
Screen Comprehension

Voice Assistant

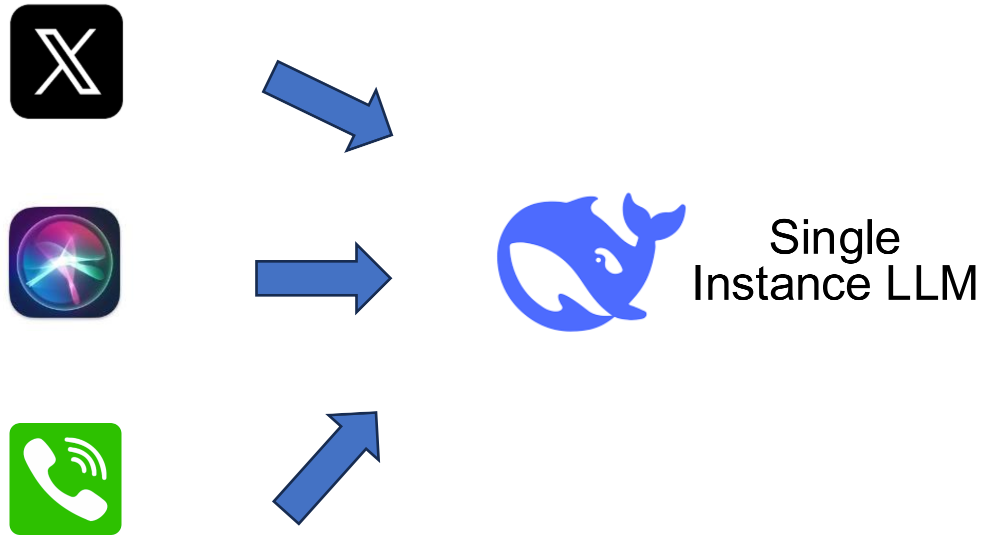
Privacy Sensitive!

Needs On-Device Deploying LLMs!

A Paradigm Shift

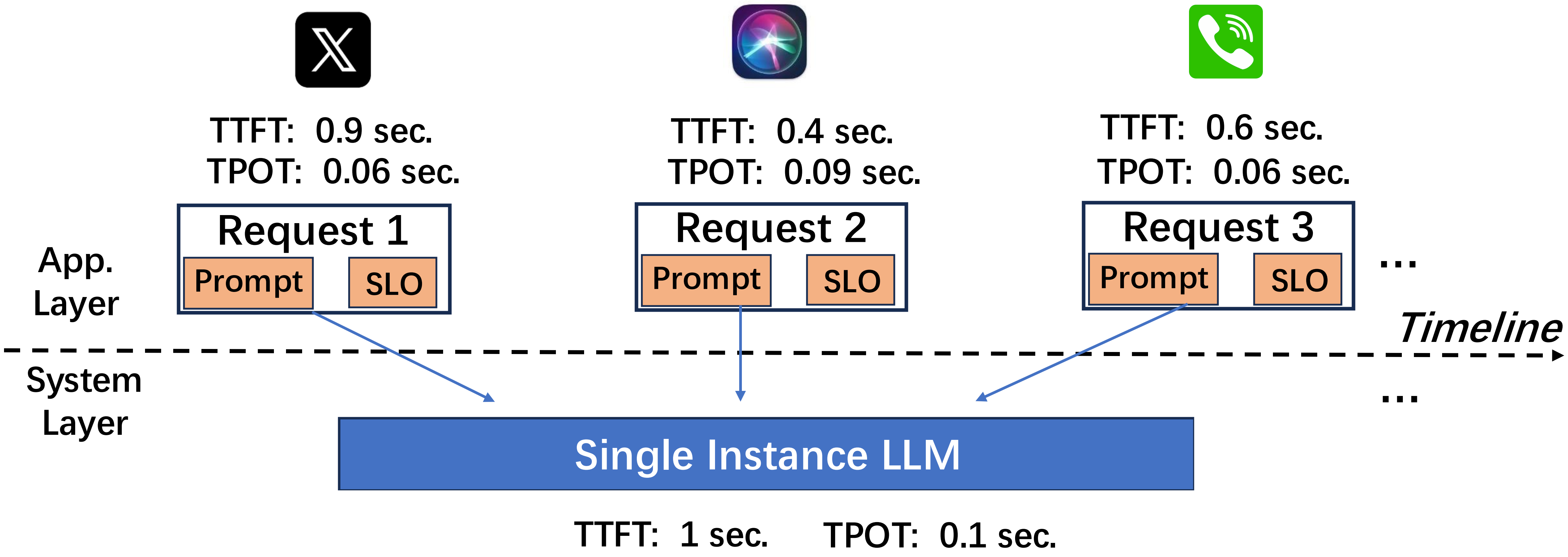


Previous: task-specific DNNs



Now: single-instance LLM service

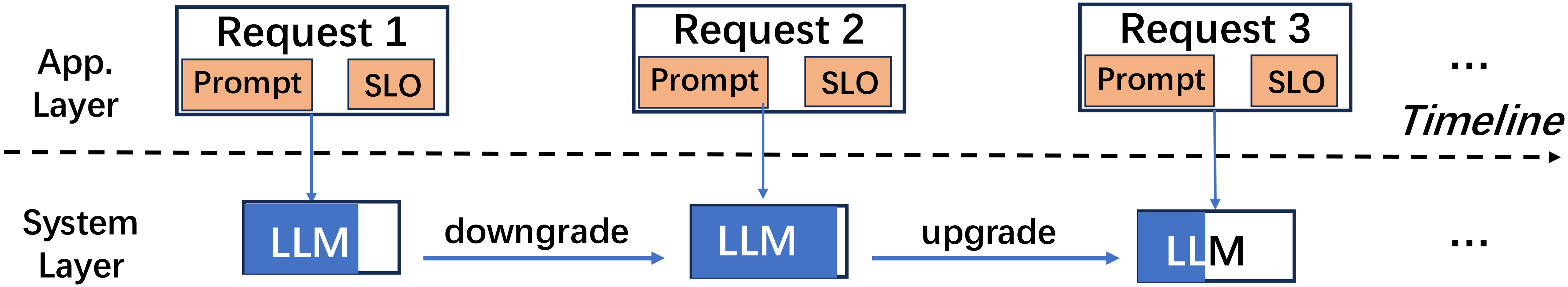
The Problem



TTFT: Time-To-First-Token *TPOT: Time-Per-Output-Token* *SLO: <TTFT, TPOT>*

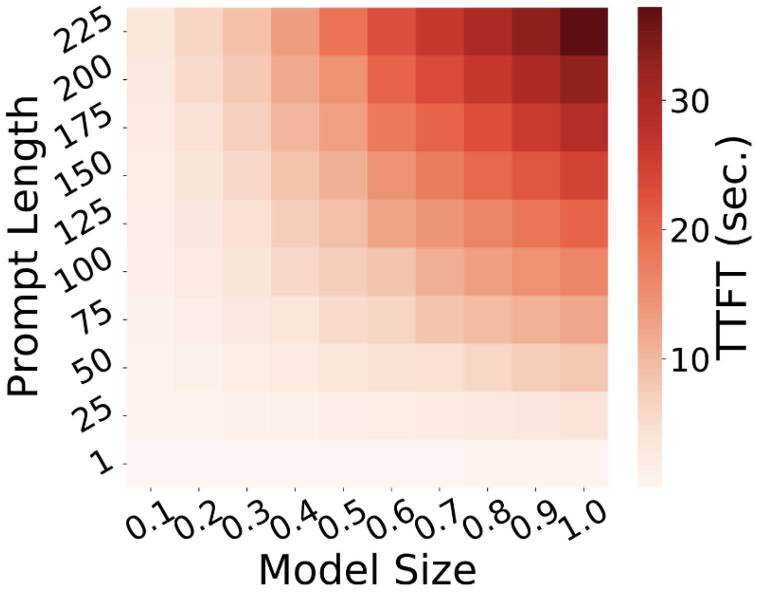
How to serve requests with various SLOs by a single LLM?

Our System

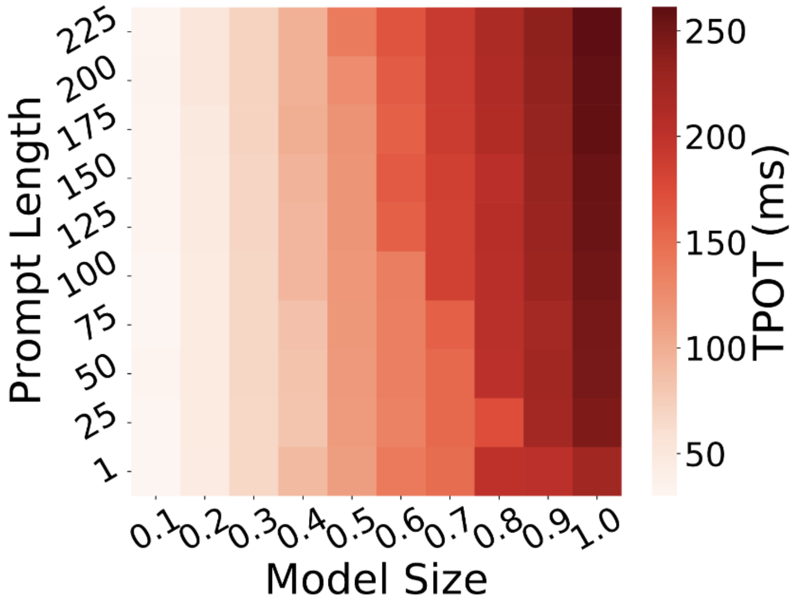


We present* ElastiLM*, a runtime elastic on-device LLM service.

Problem Analysis



(1) Prefill Latency



(2) Decode Latency

LLM inference latency w.r.t. prompt length and model size.
Measured on LLaMA-7B, Redmi K60 Champion (Snapdragon 8Gen2).

The LLM inference latency is jointly impacted by model size and prompt length.

Problem Analysis

Elasticizing the model

LLM Inference

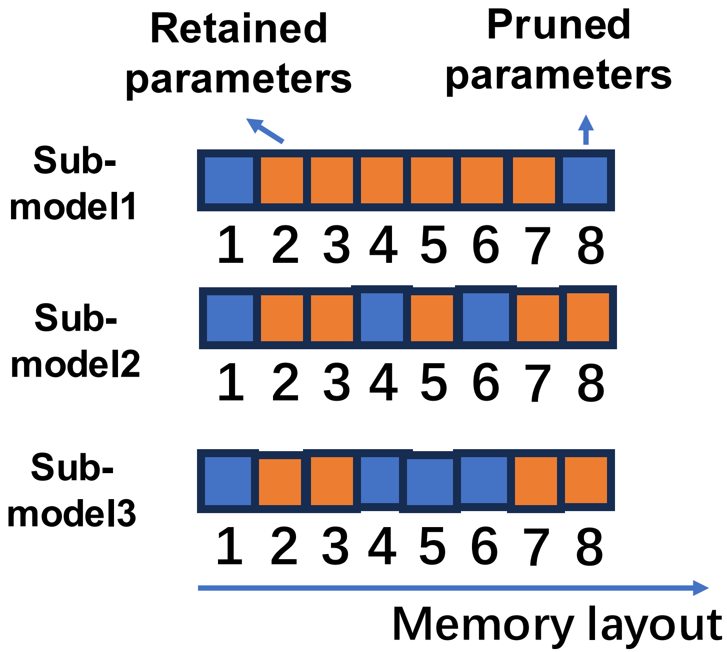
Elasticizing the prompt

Pruning into sub-models

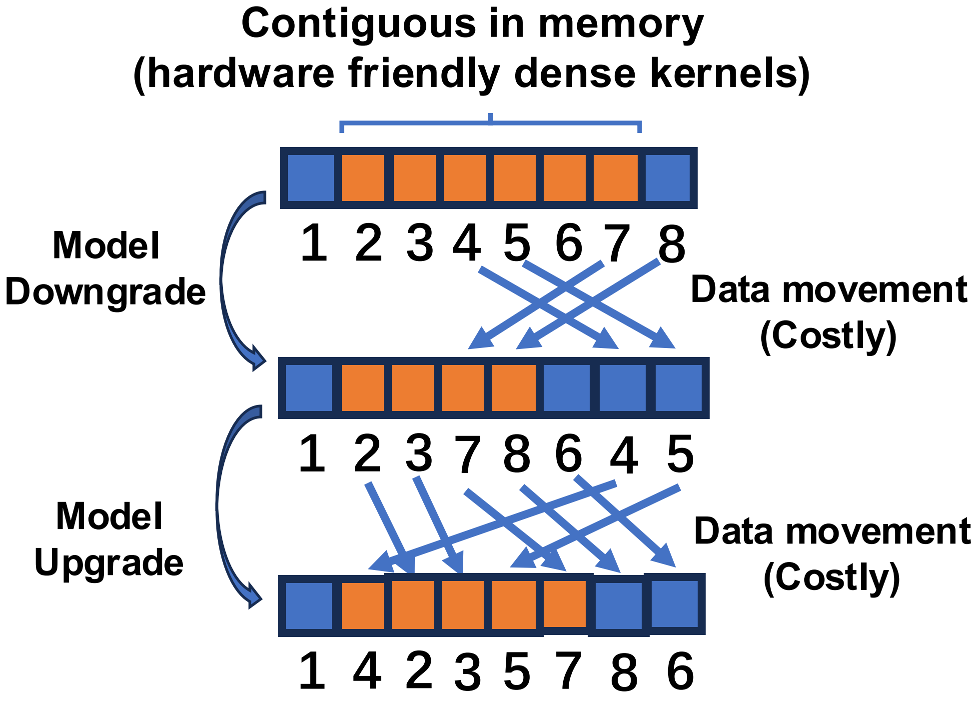
Filtering tokens by lightweight model

Consider the two dimensions for elasticity (this work)

Challenge#1



(1) Sub-models.

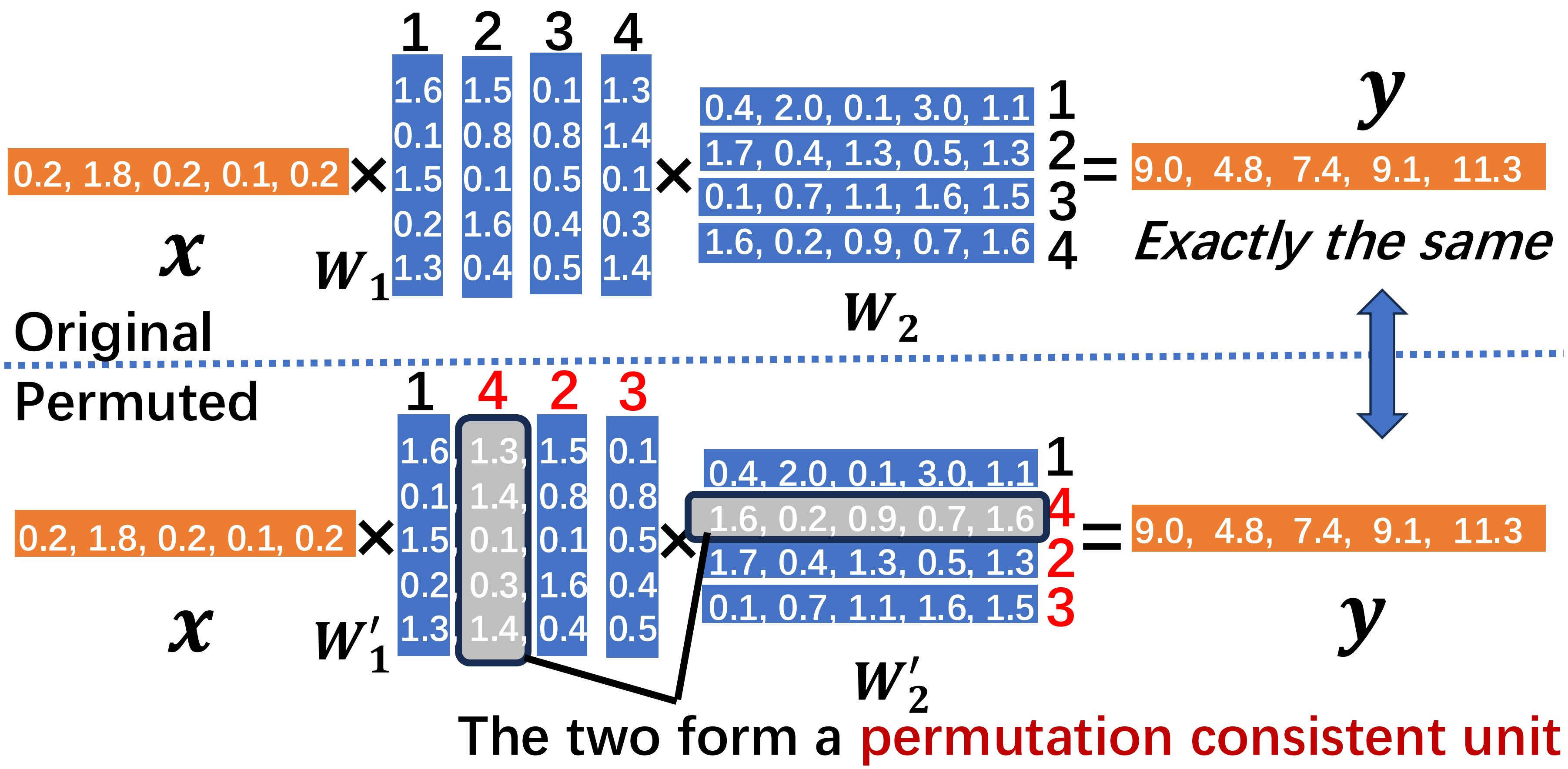


(2) Model upgrade/downgrade.

Movement of LLaMA-7B's a W_Q matrix (4096×4096) takes 139 ms on Redmi K60 Champion smartphone in the worst case.

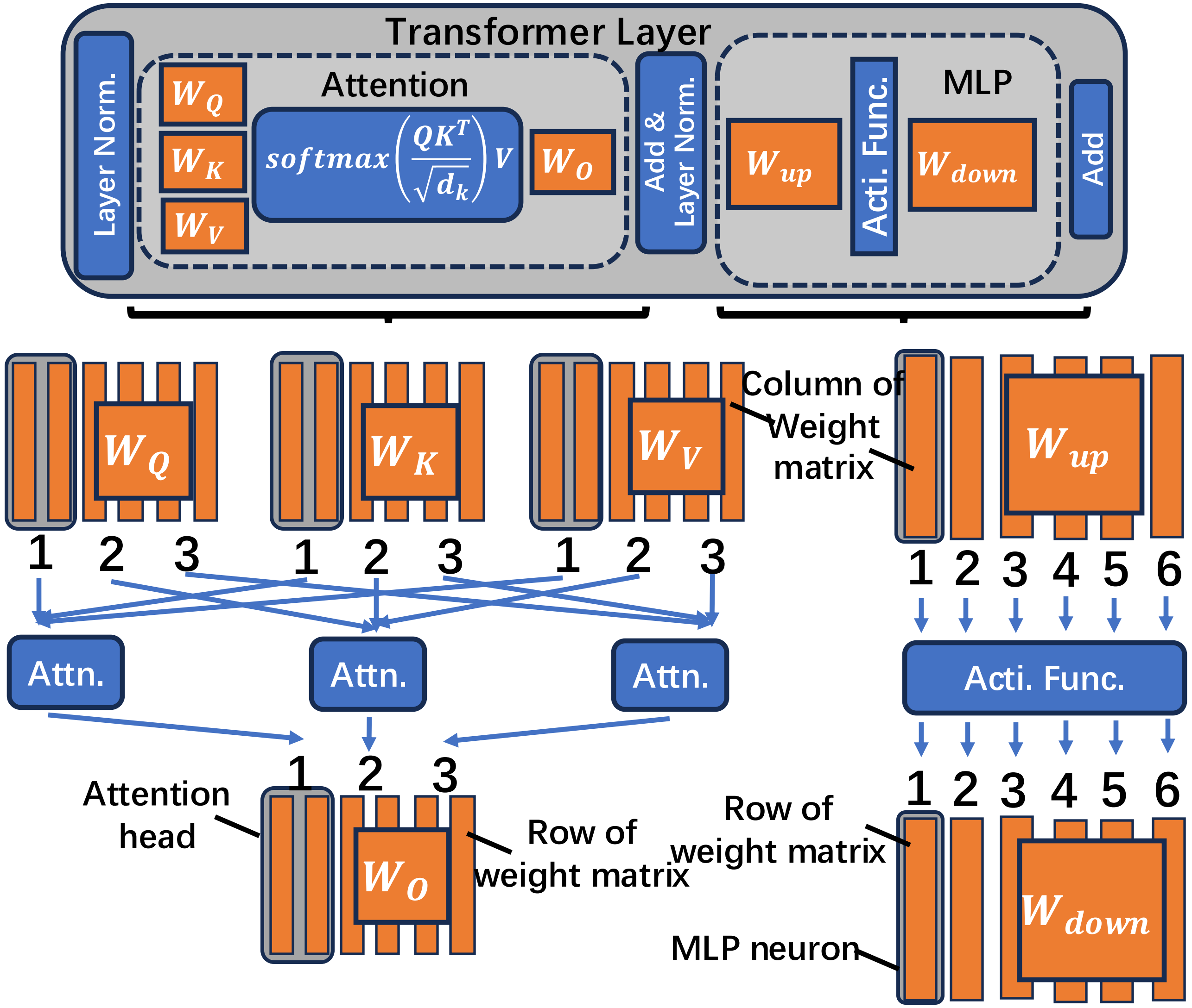
The switching of sub-models can be costly.

Insight: the permutation consistency



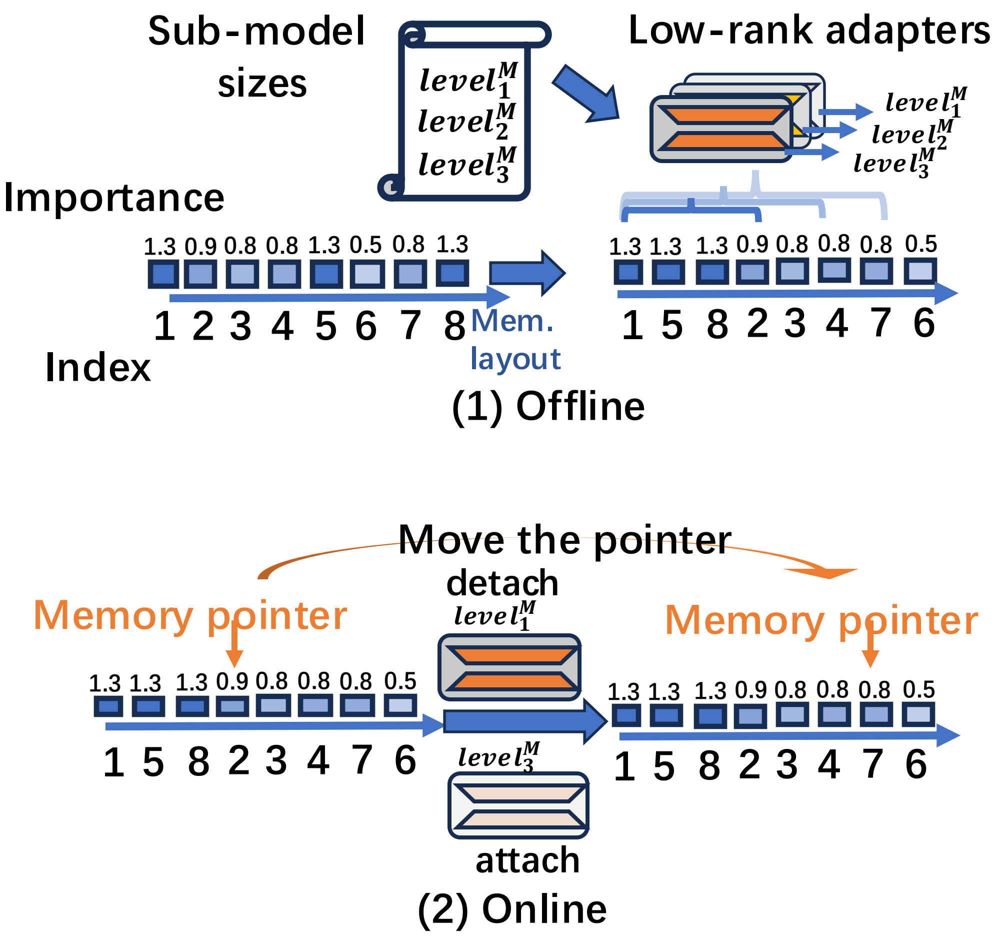
Insight: the permutation consistency

*Each **head** is a permutation consistent unit.*



*Each **neuron** is a permutation consistent unit.*

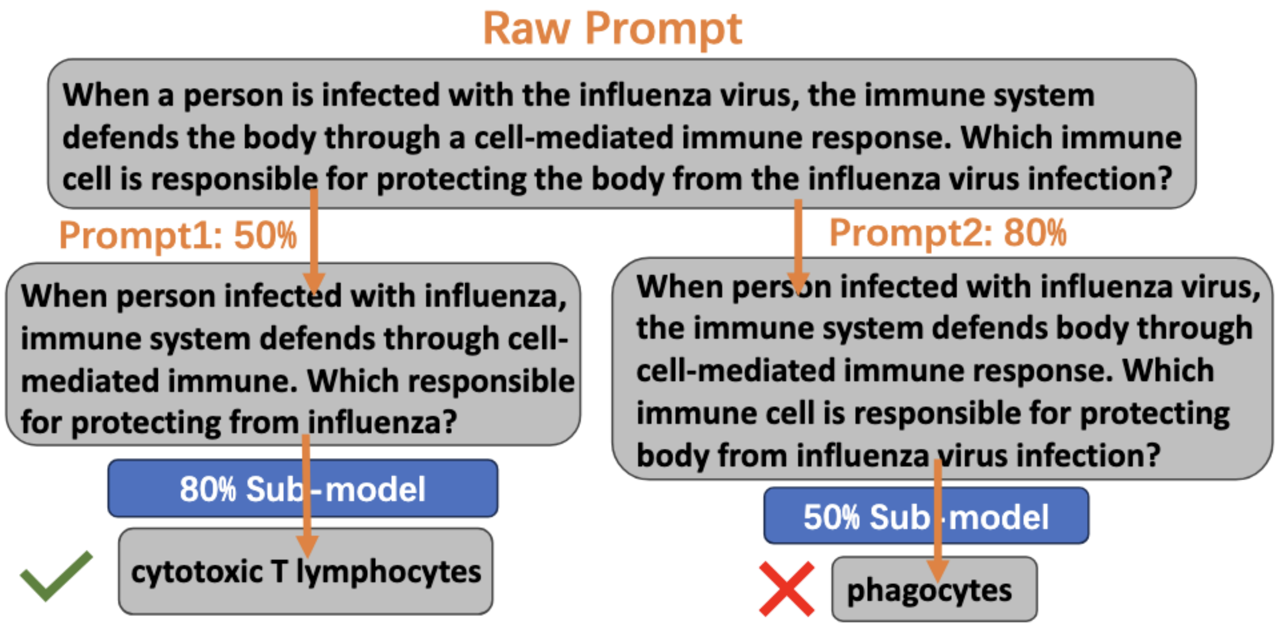
Design of Model Elastification



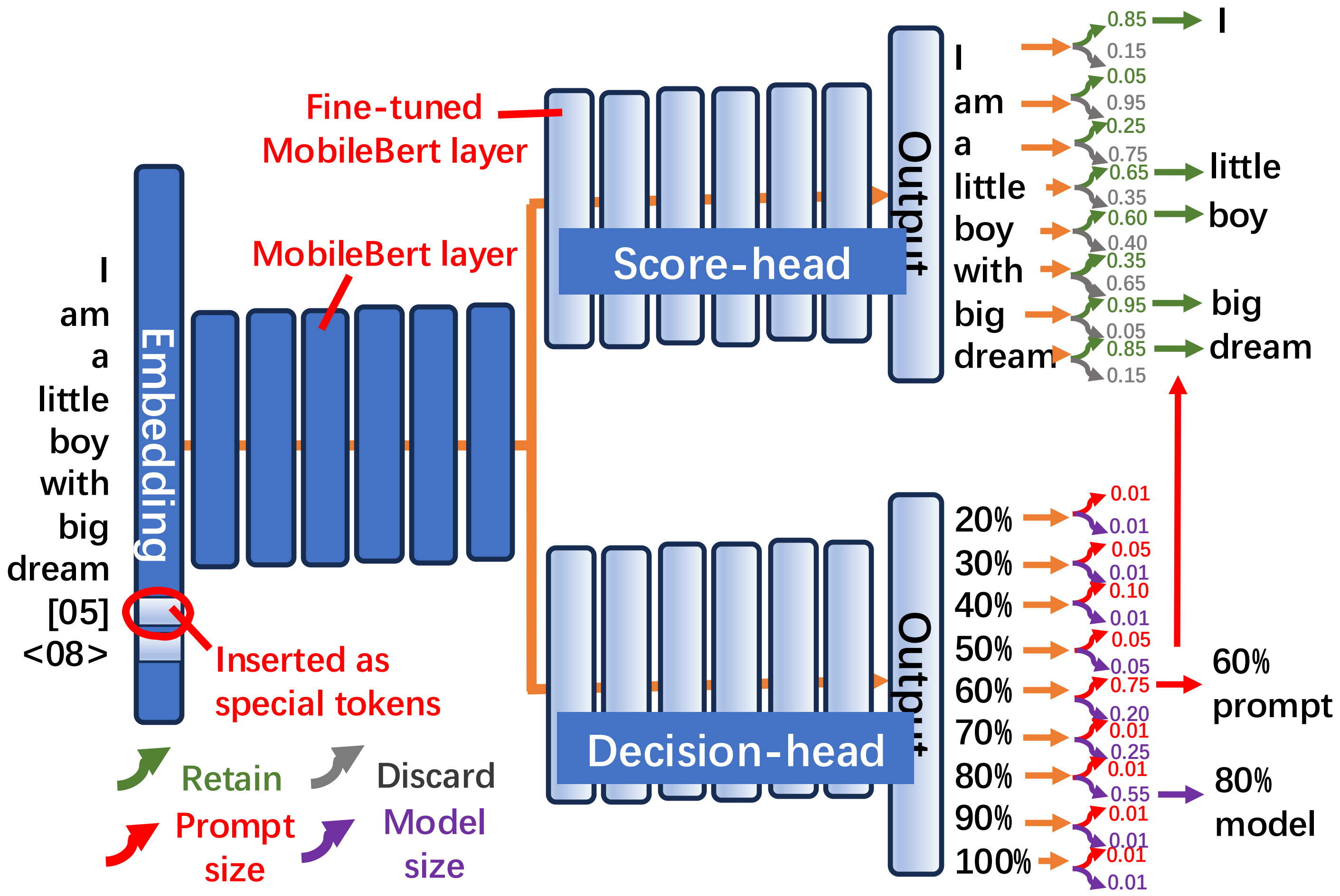
- One-shot unit-reordering
- Profiling unit importance through explainable AI
- Task-agnostic low-rank recovery of sub-models

Challenge#2

The prompt elastification and model elastification need careful orchestration.



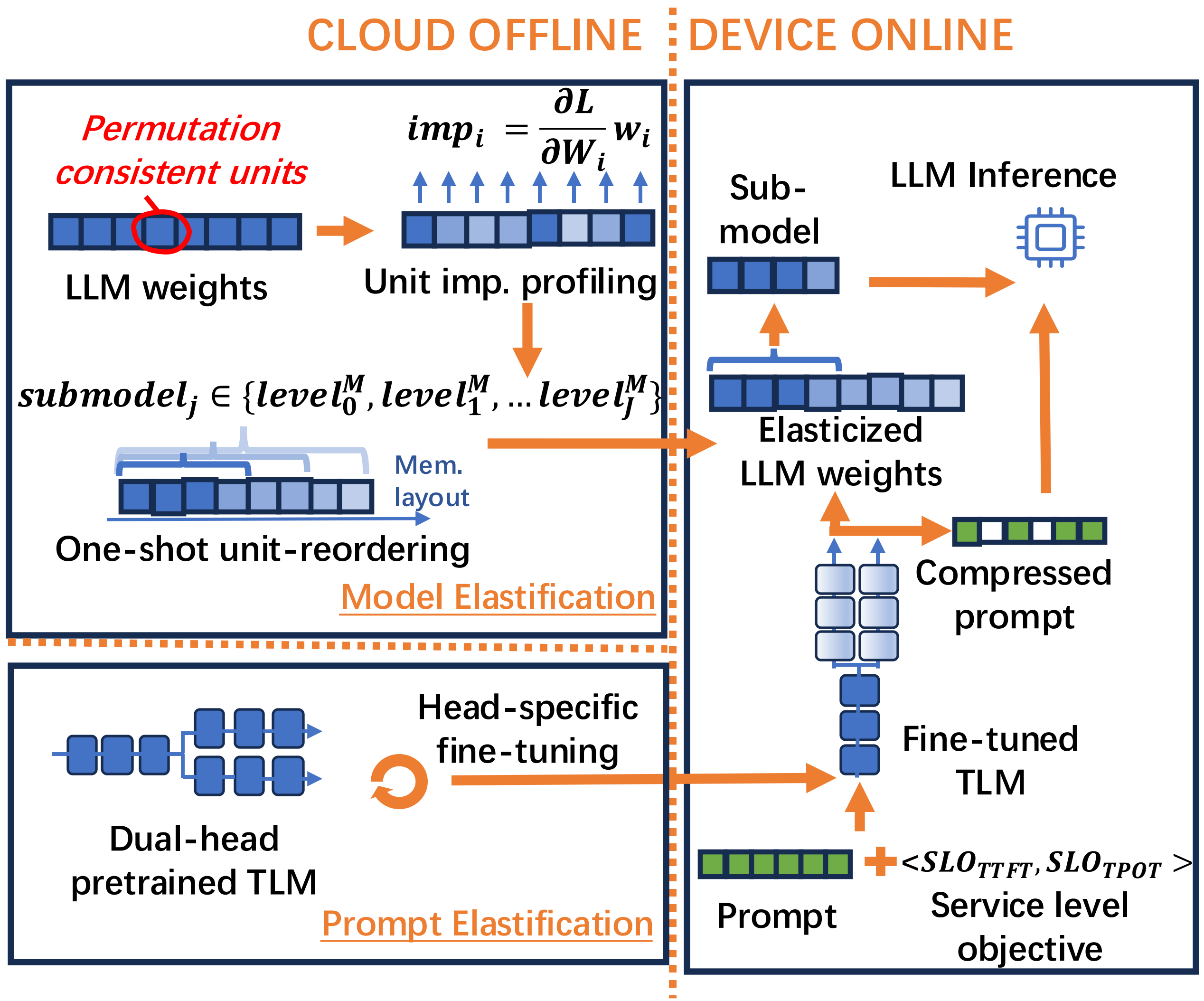
Design of Prompt Elastification



- Dual head tiny language model for prompt elastification and prompt-model orchestration
- Self-induced labelling process for decision-head training

Please refer to our paper for technique details!

Overall Workflow



Implementation and Evaluation Setup

- ***System Implementation***
  - **Based on open source framework mllm**
  - **Replacing Linear operator by ElasticLinear**
    - Adding extra memory pointer on top of the original dense kernel
  - **Optimizing the low-rank add and multiplication by ARM NEON**

- ***Test Platform***
  - **Cloud Server**
    - 64-core CPU (750GB RAM) and 8 A40 GPUs (45GB HBM each)
  - **Smartphones**
    - Redmi K60 Champion (Snapdragon 8Gen2), MI14 (8Gen3), K70 Pro (8Gen3)

Implementation and Evaluation Setup

- ***Models***
  - **Base model**
    - LLaMA-7B
    - Llama3-8B
    - Orca-mini-3B
  - **Instruction following model**
    - Vicuna-V1.5-7B
    - Llama3-instruct-8B

- ***Datasets***
  - **ARC_E**
    - Scientific QA and knowledge reasoning
  - **OBQA**
    - Common sense reasoning
  - **Octopus**
    - System API calling
  - **LlamaTouch**
    - UI Automation
  - **E2E synthesized traces**

- ***SLOs***
  - *Random set 6 levels of SLOS, based on stepwise sensitivity hierarchy.*

Apps	SLO	Dataset
Rewind	<100%, 100%>	OBQA
GMail	<80%, 90%>	ARC_E
Octopus	<60%, 80%>	Octopus

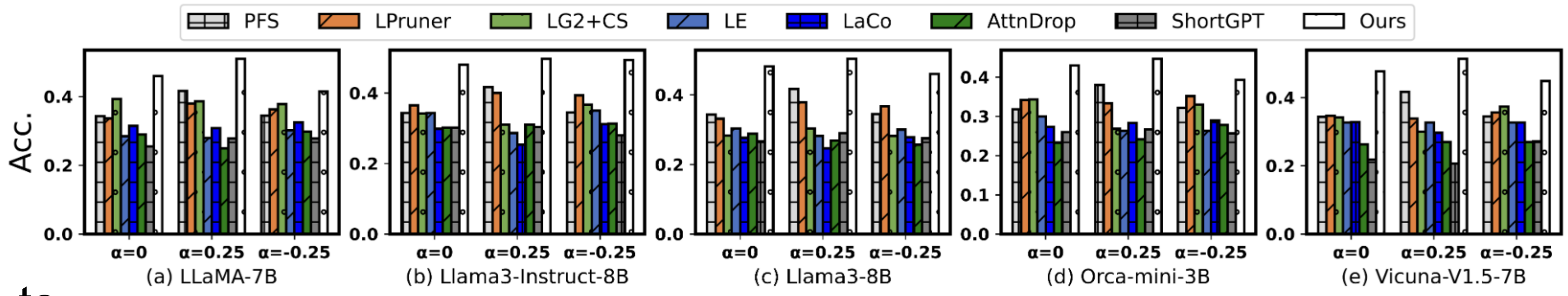
Apps	SLO	Dataset
Shortcuts	<40%, 70%>	LTouch
Gboard	<20%, 60%>	OBQA
XiaoAi	<20%, 50%>	ARC_E

Implementation and Evaluation Setup

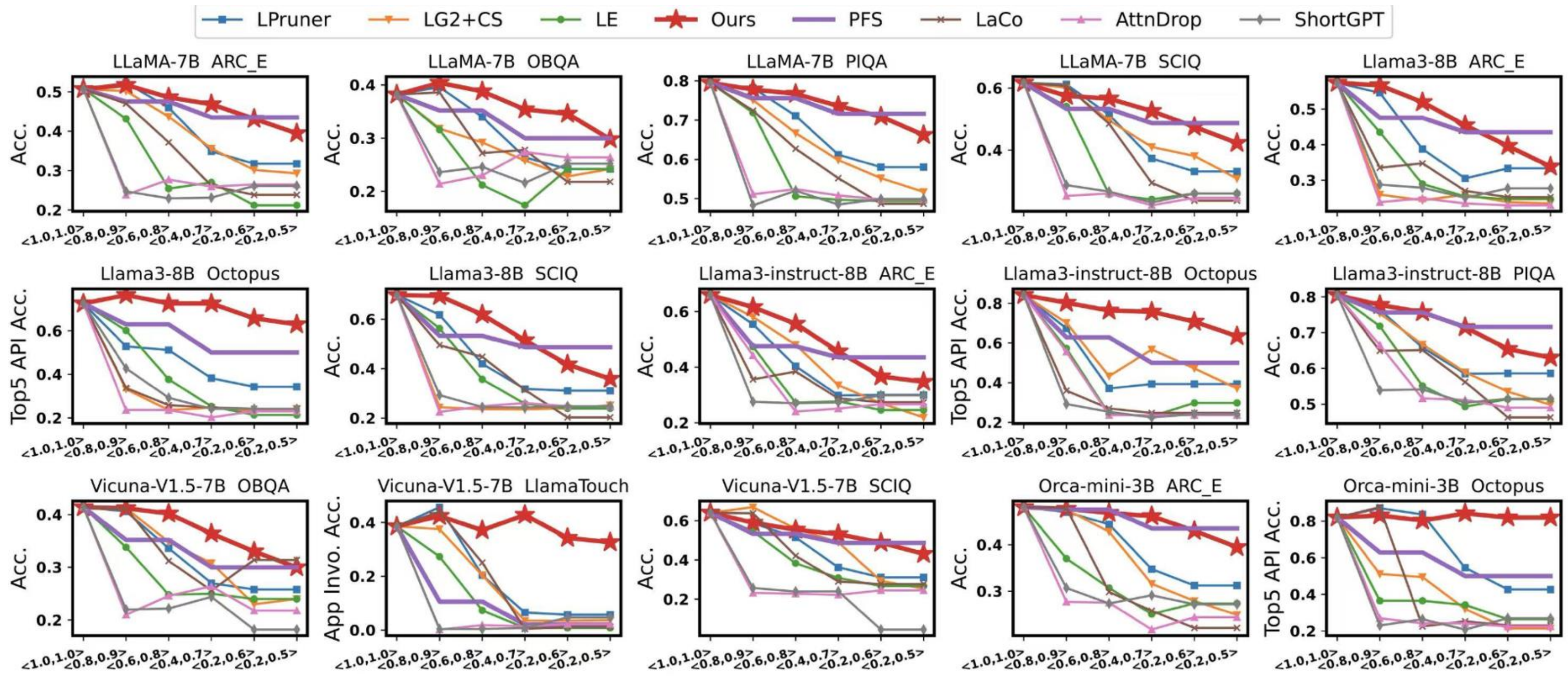
- ***Baselines***
  - **Pretraining from scratch (PFS)**
    - Directly using off-the-shelf pretrained models with various sizes (The OPT family)
  - **LLMPruner (LPruner)**
    - State-of-the-art parameter pruning method for model elastification
  - **Layerwise Elastification (LE)**
    - Pruning in a granularity of a layer
  - **LLMLingua2 + Contextual Sparsity (LG2+CS)**
    - Compress the prompt text at prefill stage, dynamically activate the neurons at decode stage
  - **LaCo/ShortGPT/AttnDop**
    - Other fine-grained model elastification methods

Evaluation Results

With meeting all SLOs, on end-to end traces, our system achieves up to 16.83%, on average 11.04% absolute accuracy improvement; our system also achieves up to 34% and on average 22% absolute accuracy improvement on standalone datasets.



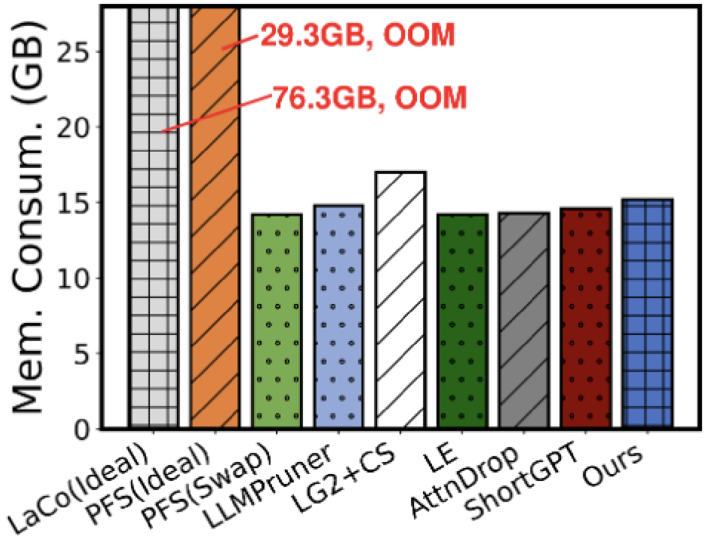
End-to-end request response accuracy on the traces.



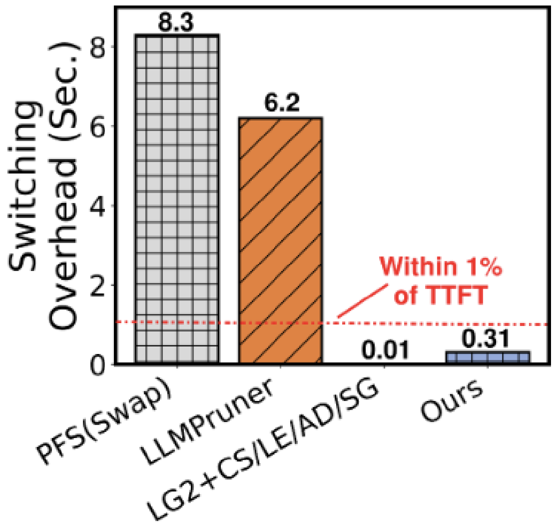
Performance under one SLO on an entire standalone dataset.

Evaluation Results

Compared to baselines, our system does not introduce obvious extra memory consumption; the switching overhead between various models is within 1% of TTFT.



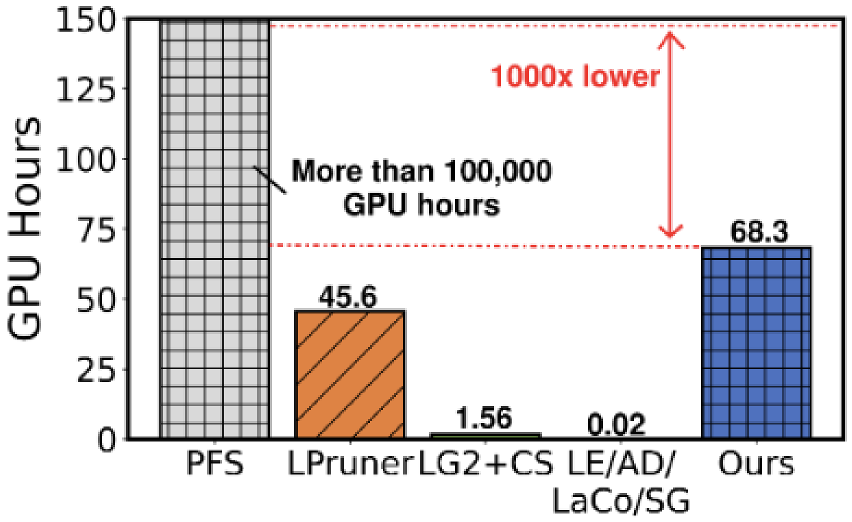
(1) Memory consumption



(2) Switching time

Evaluation Results

The offline overhead of elastification is only 68.3 GPU hours (cost about 100 dollars for cloud GPU renting), being affordable for most of developers.



Corpus Source	# of Tokens	GPU Hours
Bookcorpus	6.4K	0.03
Alpaca-cleaned	50M	40.4
Meetingbank	50M	1.7
MMLU-Pro	340K	26.3

Take Aways

- We present ElastiLM, an **elastic** on-device LLM service for serving requests with **various SLOs.**
- ElastiLM innovatively elasticizes a single LLM through both the **model dimension** and the **prompt dimension**.
- ElastiLM significantly outperforms competitive baselines by **up to 14.83% and 10.45% on average** in accuracy.

Code: https://github.com/yinwangsong/ElastiLM

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